Taheer Khan

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Education

Toronto Metropolitan University — Expected 2028
Computer Engineering — *Toronto, ON*

Relevant Courses: Embedded Systems, Digital Systems, Control Systems, Microprocessor Systems, Circuits I/II/III, Signals & Systems I/II, Operating Systems, System-On-Chip Design, Low Power Digital Integrated Circuits

Experience

Electrical Co-Lead — Sept 2024 – Current
TMU Baja Racing SAE — *Toronto, ON*

- Designed the vehicle's electrical harnesses from battery & fuse box, integrating power distribution & sensor systems.
- Weatherproofed harnesses with sleeves & heat-shrinking; **tested** for **automotive grade reliability standards**.
- Designed a **custom PCB** for gear sensor with **thermal management** and **LCD** for real-time gear display.
- **Mentored team members** on harness assembly, PCB design, and best practices for electrical integration.
- Currently working on **CAN** communication along with brand new sensors & a driver dash with a central architecture

Hardware Intern — May 2025 – Aug 2025
Aerosports — *Oakville, ON*

- Programmed **ESP32** microcontrollers using **MicroPython** to add **NFC** peripherals into embedded arcade systems.
- Helped design and assemble **custom PCBs** to interface card readers with arcade control circuitry using **datasheets**.
- Modified embedded hardware to support **UART** communication and system-level compatibility.
- Collaborated with team members to **solder**, deploy, and validate NFC readers integrated with a secure backend API.
- **Tested** & validated data flow between NFC readers and PC systems to ensure reliable account & transaction handling.

Software Developer Intern — Jan 2026 – April 2026
EventSystemPro — *Toronto, ON*

- Upgraded legacy PHP codebase to PHP 8.4, improving compatibility across core platform functionality.
- Patched **25+ applications** and library files to resolve deprecated functions, removed features, and runtime errors.
- **Tested & validated** user workflows end to end identifying issues and confirming stable system behavior.
- **Debugged production issues** using PHP stack traces and server logs to to resolve critical platform failures.
- Managed version control through structured **Git branches & commits** while updating clearly with team members.

Projects

Embedded Game Console | *C, ARM Cortex-M3, Keil uVision*

- Developed a media center with a photo gallery, mp3 player, & joystick based games using **ARM Cortex-M3** MCU.
- Implemented **RTOS-based architecture** to manage GUI & USB audio with volume control, & game logic
- Debugged RTOS firmware using tools eg. **performance analysis**, **disassembly**, & watch windows to identify scheduling & latency issues.

Binary Keyboard | *C++, KiCAD, XIAO SAMD21, SolidWorks*

- Built a binary-input keyboard using **SAMD21 MCU**, with mechanical switches, & **metal control knob**.
- Developed **C++ firmware** to process & transmit USB **HID** output while decreasing phantom clicks **by 100%**
- Designed a **custom PCB & a 3D-printed case** with **hot-swap** mechanical switch hardware into a compact build.

Smart Fitness Watch | *C, C++, LTSpice, XIAO nRF52840, KiCAD, SolidWorks*

- Built a smart watch in **C/C++** with a heartrate monitor, compass, total steps, distance, & battery management.
- Designed a heart-rate sensor using a **photodiode** to measure BPM with low-noise amplification, tested in **LTspice**.
- Integrated a **9-DOF IMU** for step tracking & compass, and monitoring battery with a **low voltage BMS**.
- Developed a cyberpunk style UI on an **SPI LCD**, and designed a custom PCB and slim 3D-printed case/strap.

Technical Skills

Languages: C/C++/C#, Assembly, VHDL, Verilog, MatLab, Python, Java, JavaScript, HTML/CSS, SQL, SSH
Technologies/Tools: Cadence, Quartus, Alterra, KiCad, MultiSim, Simulink, LTSpice, GIT, Linux
Hardware: ARM Cortex-M3, FPGA, Raspberry Pi, ESP32, ATMega, STM32, OP-AMP, MOSFET, Oscilloscope, Soldering